

An Improved Initialization-based Evolutionary Algorithm for the Top- k 2-clubs Problem

Pham Dinh Thanh

Faculty of Natural Science and Technology, Tay Bac University, Vietnam
thanhpd05@gmail.com, thanhpd@utb.edu.vn

Abstract. The s -club problem has been widely applied in social network analysis and biological network studies. Among its variants, the Top- k 2-clubs problem has attracted considerable attention from the research community. The objective of this problem is to identify k large 2-clubs that maximize a combined score reflecting both their sizes and pairwise dissimilarities. This study presents an improved evolutionary algorithm to address the Top- k 2-clubs problem. The proposed approach focuses on enhancing the efficiency of the population initialization process. Specifically, it integrates both greedy and random strategies to achieve a balance between individual solution quality and population diversity. Experimental evaluations conducted on instances from the DIMACS library demonstrate that the proposed algorithm outperforms the original approach in nearly two-thirds of the test cases.

Keywords: Genetic Algorithm · Top- k 2-clubs problem · Evolutionary Algorithms · DIMACS.

1 Introduction

Determining network connectivity plays an important role in various domains. Among these, dense subgraph detection has attracted significant research attention, particularly in addressing challenges in social network analysis and biological networks. Early studies primarily focused on clique-based problems, which aim to identify subgraphs where every pair of vertices is directly connected by an edge. However, this strict requirement of direct connectivity is often unrealistic in many real-world scenarios—such as social networks—where not all individuals are directly connected to one another.

To address this limitation, the strict requirement that two vertices must be directly connected by an edge is relaxed by allowing a path of limited length between them, giving rise to the class of s -club problems.

The concept of the s -club, introduced by Mokken in 1979 [16], was proposed to explore vertex set coverage within a graph. As a fundamental mathematical model, the s -club framework facilitates research in graph-based information mining [10]. An s -club is a subgraph in which the distance between any two vertices does not exceed s . In other words, all vertices in an s -club are at most s hops away from each other.

The s -club model has found numerous applications, such as analysing protein–protein interaction networks through clustering with a minimal number of s -clubs [17]. Similar approaches have also been investigated in social network analysis [14,15,12,7]. Furthermore, the s -club concept has been applied to partition graphs into discrete clusters, each representing an s -club [8].

Several variants of the s -club problem have been studied. One of the earliest and most widely examined formulations is the *Maximum s -club problem*, which aims to identify the largest possible s -club in a graph. This problem is known to be NP-hard for all $s \geq 1$ [5,18]. Another related challenge, known as the *Maximum Disjoint k -club Covering problem* [9], involves determining up to k non-overlapping s -clubs (each containing at least two vertices) such that the collection covers the maximum number of vertices in the graph. In 2019, another important variant of the s -club problem, known as the *Top- k 2-clubs problem* (Topk2clubs), was investigated by Mauro Castelli and colleagues [6]. In their study, the authors proposed a *Genetic Algorithm* (GA) to address the Topk2clubs, demonstrating its effectiveness in identifying multiple 2-clubs within complex networks.

Evolutionary algorithms [1,11] have been successfully applied to a wide range of theoretical and practical problems [13,19]. Although the study [6] demonstrated their effectiveness in solving the Topk2clubs, it relied primarily on basic evolutionary operators. Therefore, the present research aims to enhance the performance of the algorithm by improving certain evolutionary components, with particular emphasis on the population initialization method.

In summary, the main contribution of this work lies in proposing an initialization method for the Topk2clubs, which combines greedy and random strategies to achieve a balance between population diversity and solution quality.

The remainder of this work is arranged as follows: Section 2 discusses the problem’s definitions and notations, and the related works are presented in Section 3. The proposed methods are discussed in Section 4. Section 5 contains the experimental settings, computational results on several test sets, and a performance comparison with other algorithms. Finally, Section 6 includes the conclusions and discussion of extensions.

2 Problem definition and notations

In this section, we present some definitions that will be useful in the rest of the paper.

All graphs considered in our work are undirected and simple.

Given an undirected and simple graph $G = (V, E)$. For a vertex set $V' \subseteq V$, let $G[V']$ denote the subgraph induced by V' , where E' is defined as follows: $E' = \{(u, v) : (u, v) \in E \wedge u, v \in V'\}$. Given two vertices $u, v \in V'$, the distance between u and v in $G[V']$, denoted by $d_{G[V']}(u, v)$, is defined as the minimum number of edges on a path in $G[V']$ connecting vertices u and v .

For a vertex $v \in V$, we denote by $N_G(v)$ the set of neighbors of v , i.e., $N_G(v) = \{u \mid (u, v) \in E\}$ and denote by $N_G[v]$ the close neighborhood of v , i.e., $N_G[v] = N_G(v) \cup \{v\}$.

The diameter of $G[V']$ denoted as $\text{diam}(G[V'])$ is the maximum distance between any two vertices of V' .

Definition 1 (s -Club). Let $G = (V, E)$ be an undirected, unweighted, and simple graph, and let $s \geq 1$ be an integer. For any vertex subset $U \subseteq V$, the induced subgraph $G[U]$ is called an s -club if its diameter is at most s .

Figure 1 depicts an example of the 2-club. The subgraph induced by the vertex set $V' = \{1, 2, 3, 4, 5, 6\}$ is a 2-club.

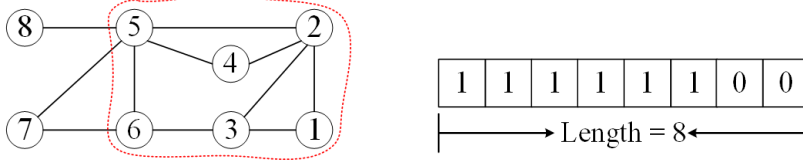


Fig. 1: An example about s -club Fig. 2: An example about individual encoding

Notice that an s -club must be a connected graph.

The **Topk2clubs** is stated as follows:

Input: a graph $G = (V, E)$, a value $\lambda \geq 0$.

Output: a set $W = \{G[W_1], \dots, G[W_k]\}$ of k 2-clubs, with $1 \leq k < |V|$ and $W_i \subseteq V$ that maximizes the following value

$$r(W) = \sum_{i=1}^k |W_i| + \lambda \sum_{i=1}^{k-1} \sum_{j=i+1}^k \text{dist}(G[W_i], G[W_j])$$

where

$$\text{dist}(G[W_i], G[W_j]) = \begin{cases} 2 - \frac{|W_i \cap W_j|^2}{|W_i||W_j|} & \text{if } W_i \neq W_j, \\ 0 & \text{else.} \end{cases}$$

3 Related works

The study of dense subgraph discovery has been a central topic in graph theory and network analysis. Among the various formulations, the concept of a *2-club* — a subgraph with diameter at most two — has received particular attention as a relaxation of the classical clique problem. While a clique requires every pair of vertices to be directly connected, a *2-club* allows connections within two hops, making it more suitable for modeling real-world networks that often contain missing or noisy links.

Early research on *2-clubs* primarily focused on the detection of a single *2-club* with the maximum number of vertices, known as the *Maximum 2-club Problem*. Due to its NP-hardness, several heuristic and metaheuristic methods have been

developed, including greedy algorithms, local search approaches, and evolutionary algorithms [3,20,4].

To capture multiple dense and cohesive communities in a network, the [Topk2clubs](#) problem was introduced by Castelli *et al.* [6]. This problem aims to identify a set of up to k 2-clubs that together represent the most significant cohesive structures within the graph. Unlike the classical *Maximum 2-Club Problem*, which yields a single subgraph, the Top- k variant promotes diversity among the detected subgraphs by maximizing the total coverage of vertices while minimizing their overlap. In [6], the authors also demonstrated the NP-hardness of the [Topk2clubs](#) problem.

To solve this problem, Castelli *et al.* [6] proposed a [GA](#) that encodes each individual as a binary string representing vertex membership in a 2-club. Their approach incorporates specialized crossover and mutation operators to preserve feasibility and promote exploration of diverse regions of the search space.

Although the application of the [GA](#) to solve the [Topk2clubs](#) has proven to be effective, the existing approach relies solely on random binary sequence initialization, leaving considerable room for improvement. Therefore, this study focuses on enhancing the population initialization method to improve the overall performance of the algorithm.

4 Proposed Algorithm

4.1 Research Motivation and Key Issues

Evolutionary algorithms are widely applied to solve complex problems in both science and engineering. When applying an evolutionary algorithm, one of the first issues to address is how to encode a solution of the problem into a chromosome and then initialize individuals in the first population. The initialization method significantly affects the quality of the obtained solutions.

In paper [6], individuals are generated by randomly creating binary strings. Therefore, the generated individuals are not necessarily 2-clubs. In addition, the objective function of the [Topk2clubs](#) tends to increase as the size of each 2-club W_i grows. Therefore, this paper proposes an initialization method that generates 2-clubs with the largest possible sizes.

4.2 Algorithm scheme

The main steps of the proposed algorithm (referred to as [Improve Genetic Algorithm \(IGA\)](#)) is presented in Algorithm 1. Another point to note in Algorithm 1 is that each individual will encode a 2-club so a solution of the [Topk2clubs](#) will include many individuals in the population. Therefore, to solve the [Topk2clubs](#), after applying the evolutionary algorithm, it is necessary to apply the greedy algorithm to find the solution to the [Topk2clubs](#) from the individuals in the population received from the evolutionary algorithm.

The implementation steps of the proposed algorithm are discussed in detail in the following subsections.

Algorithm 1: NEW APPROACH TO SOLVE THE [TOPK2CLUBS](#)

```

1 begin
2    $P_0 \leftarrow$  Randomly generate  $N$  individuals  $\triangleright$  Refer to Algorithms 3;
3   Evaluate the fitness of individual  $ind_j$  in population  $P_0$ ;
4    $t \leftarrow 0$ ;
5   while stopping conditions are not satisfied do
6      $P'_t \leftarrow$  Tournament Selection( $P_t$ )  $\triangleright$  The population at the  $t^{th}$ 
       generation is denoted as  $P_t$ ;
7      $O_t \leftarrow$  Perform crossover and mutation( $P'_t$ )  $\triangleright$  Refer to Algorithm 4
       and 5;
8     Evaluate offspring  $c_j \in O_t, \forall j \in [1, N]$ ;
9      $R_t \leftarrow O_t \cup P_t$ ;
10     $P_{t+1} \leftarrow$  Select  $N$  fittest members from  $R_t$ ;
11     $t \leftarrow t + 1$ ;
12   $W \leftarrow$  Selection individuals in  $P_t$  for constructing a solution  $\triangleright$  Refer
    to Algorithms 2;
13  return  $W$ ;

```

Algorithm 2: SELECTION OF [TOPK2CLUBS](#)

Input: Population P ; Number of selected solutions k ;
Output: Set of top- k 2-clubs $S = \{G[S_1], G[S_2], \dots, G[S_k]\}$;

```

1 begin
2    $S \leftarrow \emptyset$   $\triangleright$  Initialize selected set;
3    $S_1 \leftarrow$  An individual with a 2-club containing the maximum number
     of vertices;
4    $W \leftarrow W \cup G[S_1]$ ;
5   for  $t \leftarrow 2$  to  $k$  do
6      $bestInd \leftarrow null$ ;  $bestSet \leftarrow null$ ;  $bestValue \leftarrow -\infty$ ;
7     foreach individual  $c$  in  $P$  do
8        $S_t \leftarrow$  vertices corresponding to individual  $c$ ;
9        $S' \leftarrow S \cup \{G[S_t]\}$ ;
10       $value \leftarrow r(S')$   $\triangleright$  Compute objective of current selection;
11      if  $value > bestValue$  then
12         $bestValue \leftarrow value$ ;
13         $bestInd \leftarrow c$ ;
14         $bestSet \leftarrow S_t$ ;
15       $S \leftarrow S \cup \{G[bestSet]\}$   $\triangleright$  Add best 2-club to result set;
16       $P \leftarrow P \setminus \{bestInd\}$   $\triangleright$  Remove it from population;
17 return  $S$ 

```

4.3 Individual Representation

Let $G[V']$ be a 2-club of the graph $G = (V, E)$, where $V' \subseteq V$. An individual is represented as a binary array whose length equals the number of vertices in the

graph. Each gene in the individual takes a value of 1 if the corresponding vertex belongs to V' , and 0 otherwise.

Figure 2 illustrates the individual representation corresponding to the two 2-clubs shown in Figure 1. In the figures, vertices 1, 2, 3, 4, 5, and 6 belong to the 2-club; therefore, the first to sixth genes are assigned a value of 1, while the seventh and eighth genes are assigned a value of 0.

4.4 Individual Initialization

In this study, each individual represents a potential 2-club in the graph $G(V, E)$. The initialization process aims to generate high-quality individuals that are likely to form valid 2-clubs while maintaining population diversity.

Algorithm 3: INITIALIZATION OF A 2-CLUB INDIVIDUAL

Input: Graph $G(V, E)$.
Output: Binary-encoded individual representing a 2-club.

```

1 begin
2    $V_{uncover} \leftarrow V$ ;  $S \leftarrow \emptyset$ ;  $radius \leftarrow 0$ ;
3   Select a random vertex  $v_0 \in V_{uncover}$ ;
4   Add  $v_0$  to  $S$  and remove it from  $V_{uncover}$ ;
5   while  $V_{uncover} \neq \emptyset$  and  $radius < 2$  do
6     foreach  $u \in S$  do
7        $deg_u \leftarrow$  The number of uncovered neighbors of  $u$  in  $V_{uncover}$ ;
8     Find  $u^* = \arg \max_{u \in S} deg_u$ ;
9      $L \leftarrow$  all vertices in  $S$  having  $deg_u = deg_{u^*}$ ;
10    Select randomly a vertex  $u_r$  from  $L$ ;
11    Add  $u_r$  and all its neighbors  $N(u_r)$  to  $S$ ;
12    Remove  $\{u_r\} \cup N(u_r)$  from  $V_{uncover}$ ;
13     $radius \leftarrow radius + 1$ ;
14  foreach  $v \in S$  do  $solution[v] \leftarrow 1$ ;
15  return  $S$ 
```

The initialization procedure begins by randomly selecting a vertex as the seed of a candidate 2-club. The algorithm then iteratively expands the current set by adding neighboring vertices that maximize the number of uncovered adjacent vertices, ensuring that each expansion step contributes to forming a dense subgraph.

This random-greedy mechanism balances *exploration* (through random selection) and *exploitation* (through greedy expansion), ensuring that each initialized individual tends to form a 2-club with the maximum possible number of vertices. The algorithm terminates when no additional vertices is selected to expand the current 2-club.

In Algorithm 3, in addition to randomly selecting a starting vertex, the algorithm also employs a random selection mechanism when multiple candidate vertices have the same number of uncovered adjacent vertices.

4.5 Crossover operator

The hybrid operator used in this study was introduced in [6]. It combines two crossover operations, namely *AND_OR_Crossover* and *One_Point_Crossover*, applied with equal probability. The main step of crossover operator is presented in Algorithm 4.

Algorithm 4: CROSSOVER OPERATOR

Input: Parent individuals P_1, P_2 ; Number of genes n .

Output: Two offspring individuals C_1, C_2

```

1 begin
2    $r \leftarrow \text{random}()$   $\triangleright$  Randomly choose a crossover type;
3   if  $r < 0.5$  then  $C_1, C_2 \leftarrow \text{AND\_OR\_Crossover}(P_1, P_2, n, \text{rnd})$  ;
4   else  $C_1, C_2 \leftarrow \text{One\_Point\_Crossover}(P_1, P_2, n, \text{rnd})$  ;
5   return Offspring  $C_1, C_2$ 

```

4.6 Mutation operator

The study uses a mutation operator [6], which combines three types of mutations in equal proportion:

1. **Mutation 1:** Select a random vertex v_0 from the set of currently selected vertices V^+ . For every other vertex $v_i \in V^+ \setminus \{v_0\}$, check whether the shortest-path distance $d_G(v_i, v_0) \leq 2$. If not, set the corresponding bit $c[i] = 0$ to remove that vertex.
2. **Mutation 2:** Select a random vertex v_0 from the set of unselected vertices V^- . If v_0 is within distance 2 of all vertices in V^+ , flip its corresponding bit $c[j_0] = 1$ to include it in the 2-club.
3. **Bit flip mutation:** Randomly flip one or more bits in the chromosome.

The main steps of the mutation is presented in Algorithm 5.

Algorithm 5: TOP-K 2-CLUBS MUTATION

Input: Adjacency matrix E ; parent chromosome P ; number of genes n .

Output: Mutated offspring C .

```

1 begin
2   Generate a random number  $r \in [0, 1]$ ;
3   if  $r < 0.33$  then  $C \leftarrow \text{BITFLIPMUTATION}(P, n)$  ;
4   else if  $r < 0.66$  then  $C \leftarrow \text{MUTATION1}(E, P, n)$  ;
5   else  $C \leftarrow \text{MUTATION2}(E, P, n)$  ;
6   return  $C$ ;

```

4.7 Evaluate solution

The fitness function using in IGA is designed to evaluate the potential contribution of each individual to the overall solution. Since each individual represents

a candidate 2-club rather than the entire [Topk2clubs](#) solution, the fitness value must reflect how well the 2-club satisfies the problem’s structural properties. Specifically, the fitness function estimates the quality of an individual based on the number of vertices belonging to the 2-club represented by that individual, while indirectly encouraging the formation of larger and more cohesive sub-graphs. This evaluation allows the algorithm to guide the evolutionary search toward individuals that are more likely to form high-quality 2-club combinations in subsequent generations.

Given an individual ind and the input graph $G = (V, E)$, the fitness function is defined as follows [6]:

$$f(n_v, diam, S) = \begin{cases} n_v & \text{if } 0 \leq diam \leq 2; \\ \frac{1}{n_v} & \text{if } 2 < diam. \end{cases} \quad (1)$$

where n_v is the number of vertices of $G[ind]$.

5 Computational results

5.1 Experimental criteria

Criteria for assessing the quality of the algorithm outputs include the following: Average function value over all (Avg); Best function value achieved over all runs (BF); Standard deviation (STD).

5.2 Experimental Settings

The [IGA](#) is compared with the genetic algorithm that was employed in [6] (referred to as [BGA](#)) to address the same problem.

To ensure a fair comparison, all algorithms were independently executed 20 times on a machine equipped with an Intel Core i7-12700K CPU and 32 GB of RAM running Microsoft Windows 10. Both the [IGA](#) and [BGA](#) methods employed a population size of 100 individuals and performed 50,000 evaluations. The crossover and mutation rates were set to 0.9 and 0.1, respectively. All implementations were developed in the C# programming language.

5.3 Problem instances

To evaluate the performance of the proposed algorithm, instances from the DIMACS library [2] are used. The selected cases contain fewer than 1500 vertices, making them suitable for computational experimentation. Descriptive statistics for these instances are provided in Tables 1, where $|V|$ denotes the number of vertices, $|E|$ denotes the number of edges, and D_G denotes the graph density.

Table 1: Summary information of instances

Instances	$ V $	$ E $	D_G	Instances	$ V $	$ E $	D_G
adjnoun	112	425	0.068	polbooks	1490	16715	0.015
celegansneural	297	2148	0.049	johnson8-2-4	28	210	0.56
celegans_metabolic	453	2025	0.020	hamming6-4	64	704	0.35
chesapeake	39	170	0.229	c-fat200-1	200	1534	0.08
dolphins	62	159	0.084	hamming6-2	64	1824	0.90
email	1133	5451	0.009	johnson8-4-4	70	1855	0.77
football	115	613	0.094	c-fat200-2	200	3235	0.16
jazz	198	2742	0.141	C125.9	125	6963	0.90
karate	34	78	0.139	c-fat200-5	200	8473	0.43
lesmis	77	254	0.087	keller4	171	9435	0.65
hamming8-2	256	31616	0.97	brock200_3	200	12048	0.61
sanr200_0.7	200	13868	0.70				

$|V|$: The number of vertices; $|E|$: The number of edges; D_G : The density of a graph.

5.4 Experimental results

The results obtained by the algorithms are presented in Table 3. In each column, bold values indicate the instances where the proposed algorithm outperforms the corresponding algorithm, while bold and italic values represent the instances where the two algorithms perform comparably.

Table 2: Summary of the Comparison between IGA and BGA

BF			Avg		
Better	Equal	Worse	Better	Equal	Worse
12	10	1	19	0	4

The Table 2 presents a comparative summary of IGA against BGA. The columns *Worse*, *Better*, and *Equal* indicate the number of instances in which IGA performed worse, better, or equal to the compared algorithm.

The results presented in Table 2 indicate that IGA outperforms BGA in most test cases for both *BF* and *Avg* criteria. Specifically, for the *BF* criterion, IGA performs better than BGA in 12 out of 23 test cases, while for the *Avg* criterion, IGA performs better in 19 out of 23 test cases.

In summary, IGA generally demonstrates superior performance compared to BGA, consistently achieving more favorable results across most test cases.

Table 3: Results obtained by IGA and BGA on the instances.

Instances	IGA			BGA		
	BF	Avg	STD	BF	Avg	STD
adjnoun	93.0	135.1	23.0	39.0	102.4	37.8
celegansneural	696.0	696.3	0.7	852.0	852.1	0.3
celegans_metabolic	198.0	491.5	215.6	66.0	373.5	244.2
chesapeake	90.0	90.4	0.6	48.0	84.7	13.3
dolphins	31.4	38.7	3.2	20.1	33.4	4.9
email	53.1	126.8	54.8	14.6	71.4	51.5
football	36.2	41.7	4.1	28.2	40.0	5.0
jazz	279.0	307.9	7.0	137.0	260.4	50.2
karate	33.0	49.3	6.8	18.0	44.0	12.4
lesmis	144.0	149.8	9.5	126.0	188.0	15.4
polbooks	69.1	76.7	4.0	18.3	67.2	20.7
johnson8-2-4	81.0	82.0	1.0	81.0	81.8	0.9
hamming6-4	189.0	190.0	1.0	189.0	190.0	0.9
c-fat200-1	45.0	52.7	5.0	45.0	50.4	4.2
hamming6-2	189.0	190.0	1.0	189.0	190.0	0.9
johnson8-4-4	207.0	208.0	1.0	207.0	207.9	0.9
c-fat200-2	101.0	102.9	1.3	99.0	102.1	2.2
C125.9	372.0	373.0	1.0	372.0	372.9	1.0
c-fat200-5	253.9	259.0	3.3	252.0	256.8	3.8
keller4	510.0	510.9	1.0	510.0	511.0	0.9
brock200_3	597.0	598.0	1.0	597.0	597.9	0.9
sanr200_0.7	597.0	597.9	1.1	597.0	597.9	0.9
hamming8-2	765.0	765.9	1.1	765.0	766.0	1.0

6 Conclusion

The [Topk2clubs](#) has been extensively studied as it plays a crucial role in understanding community structures and group interactions within social networks. In this study, we propose an improved genetic algorithm that incorporates a novel initialization method. The proposed approach combines greedy and random strategies to generate high-quality initial individuals. Experimental results on datasets from the DIMACS library demonstrate that the proposed algorithm outperforms existing approaches. In future work, we will focus on improving individual representation and enhancing the performance of evolutionary operators for the [Topk2clubs](#).

References

1. Bäck, T., Fogel, D.B., Michalewicz, Z.: Evolutionary computation 1: Basic algorithms and operators. CRC press (2018)
2. Bader, D.A., Meyerhenke, H., Sanders, P., Wagner, D.: Graph Partitioning and Graph Clustering. American Mathematical Society (Jan 2013)
3. Balasundaram, B., Pajouh, F., Butenko, S.: Clique relaxations in social network analysis. *Optimization Letters* (2011)
4. Bourjolly, J.M., et al.: Heuristic and exact approaches for the 2-club problem. *Discrete Applied Mathematics* (2019)
5. Bourjolly, J.M., Laporte, G., Pesant, G.: An exact algorithm for the maximum k -club problem in an undirected graph. *European Journal of Operational Research* **138**(1), 21–28 (2002)
6. Castelli, M., Dondi, R., Manzoni, S., Mauri, G., Zoppis, I.: Top k 2-Clubs in a Network: A Genetic Algorithm, p. 656–663. Springer International Publishing (2019)
7. Cavique, L., Mendes, A.B., Santos, J.M.: An algorithm to discover the k -clique cover in networks. In: Progress in Artificial Intelligence: 14th Portuguese Conference on Artificial Intelligence, EPIA 2009, Aveiro, Portugal, October 12–15, 2009. Proceedings 14. pp. 363–373. Springer (2009)
8. Chakraborty, D., Chandran, L.S., Padinhatteeri, S., Pillai, R.R.: Algorithms and complexity of s -club cluster vertex deletion. In: International Workshop on Combinatorial Algorithms. pp. 152–164. Springer (2021)
9. Dondi, R., Mauri, G., Zoppis, I.: On the tractability of finding disjoint clubs in a network. *Theoretical Computer Science* **777**, 243–251 (2019)
10. Fortunato, S.: Community detection in graphs. *Physics reports* **486**(3–5), 75–174 (2010)
11. Goldberg, D.E.: Genetic algorithms. Pearson Education India (2006)
12. Gupta, P., Arora, M., Thakur, H.K.: Community detection in social networks: A deep learning approach using autoencoders. *Informatica* **49**(5) (Jan 2025)
13. Hanh, P.T.H., Thanh, P.D., Binh, H.T.T.: Evolutionary algorithm and multifactorial evolutionary algorithm on clustered shortest-path tree problem. *Information Sciences* **553**, 280–304 (Apr 2021)
14. Laan, S., Marx, M., Mokken, R.J.: Close communities in social networks: boroughs and 2-clubs. *Social Network Analysis and Mining* **6**, 1–16 (2016)
15. Mokken, R.J., Heemskerk, E.M., Laan, S.: Close communication and 2-clubs in corporate networks: Europe 2010. *Social Network Analysis and Mining* **6**, 1–19 (2016)
16. Mokken, R.J., et al.: Cliques, clubs and clans. *Quality & Quantity* **13**(2), 161–173 (1979)
17. Pasupuleti, S.: Detection of protein complexes in protein interaction networks using n -clubs. In: Evolutionary Computation, Machine Learning and Data Mining in Bioinformatics: 6th European Conference, EvoBIO 2008, Naples, Italy, March 26–28, 2008. Proceedings 6. pp. 153–164. Springer (2008)
18. Shahinpour, S., Butenko, S.: Algorithms for the maximum k -club problem in graphs. *Journal of Combinatorial Optimization* **26**(3), 520–554 (Mar 2012)
19. Thanh, P.D., Thanh Binh, H.T., Lam, B.T.: A survey on hybridizing genetic algorithm with dynamic programming for solving the traveling salesman problem. In: 2013 International Conference on Soft Computing and Pattern Recognition (SoCPaR). p. 66–71. IEEE (Dec 2013)
20. Veremyev, A., Boginski, V.: Finding maximum 2-clubs in networks. *Computers & Operations Research* (2014)